

# Electric: From Schematic To IC Layout

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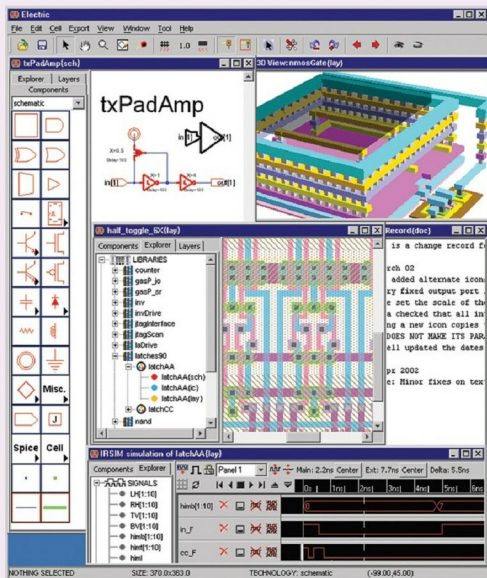
The moment you talk about electronic design automation tools, the mind jumps to the big three—Synopsys, Cadence and Mentor Graphics. While these perform your tasks more than perfectly, you need to shell out a bomb. This article takes a look at one of the popular open source tools for very large scale integration (VLSI) computer-aided design, Electric.

Electric is an EDA tool used to draw schematics and to make integrated circuit (IC) layouts. It can also handle hardware description languages like VHDL Hardware Description Language and Verilog. Throw in the various analyses options and synthesis categories, and you have a complete VLSI design package. Let us now take a glimpse into the many features of this design system.

## Technology libraries, a variety

You do not need to look at your tool's resources first, before planning your design. Most of the components you might need—be it different technologies or multiple file formats—come ready with this software package. N-channel MOSFETs and bipolar technologies based on integrated injector logic are your go-to technologies. CMOS support comes in many different versions including the six-metal double-poly MOSIS rules and CalTech's Round rules. Designs based on a hybrid of the two, the BiCMOS method, are also no hassle with Electric.

To help you ease in, each technology comes with a description of all characteristics of its



What you can do with Electric

components and wires including graphical details, design rules and simulation aspects. You just have to add in the technology module to the specific environment and use the technology editor to find your way around.

**In varied formats, too.** This tool has its own format for reading and writing libraries of circuitry. But it also ensures compatibility with other EDA systems by also

handling popular interchange and manufacturing formats like Caltech Intermediate Format (CIF), Calma interchange format (GDSII), Electronic Design Interchange Format (EDIF) and AutoCAD mechanical format (DXF). Even schematic capture packages like EAGLE, PADS, EDAD and SUE are supported. You can choose to make your plots in either HPGL or PostScript languages.